Microeconomics I — Worked solutions

Practice for the Part I exam · Class 9 · 27 October 2026 · Pavel Kocourek

The statements are in the problem list of class 9. Each part shows the steps you would write in an exam, with the reason for each step and a few checks; the results are shaded. Exact fractions throughout: no decimals are needed. Conventions as in the problem list: $u^0 = v(p^0, w)$, $u^1 = v(p^1, w)$, $\mathrm{CV} = w - e(p^1, u^0)$, $\mathrm{EV} = e(p^0, u^1) - w$ (MWG's signs).

Problem 1 · $u(x_1, x_2) = x_1\, x_2^2$

classes 4–7

(a) Demand. Shortcut: for a Cobb–Douglas utility $x_1^a x_2^b$ the consumer spends the share $a/(a+b)$ of her wealth on good 1 and $b/(a+b)$ on good 2. Here $a = 1$, $b = 2$, so the shares are $\frac{1}{3}$ and $\frac{2}{3}$:

$$p_1 x_1 = \tfrac{1}{3} w, \quad p_2 x_2 = \tfrac{2}{3} w \qquad \Longrightarrow \qquad x_1(p, w) = \frac{w}{3p_1}, \quad x_2(p, w) = \frac{2w}{3p_2}.$$

Without the rule, by the tangency condition $\mathrm{MRS} = p_1/p_2$ (it comes from the Lagrangian, as in P2(a)): $\mathrm{MRS} = \frac{\partial u/\partial x_1}{\partial u/\partial x_2} = \frac{x_2^2}{2x_1 x_2} = \frac{x_2}{2x_1} = \frac{p_1}{p_2}$ gives $p_2 x_2 = 2 p_1 x_1$; with the budget $p_1 x_1 + p_2 x_2 = w$ this is $3 p_1 x_1 = w$: the same demands.
Why this is the optimum. u is increasing, so the budget binds (money left over would buy more of a good, and more is better). At a corner ($x_1 = 0$ or $x_2 = 0$) $u = 0$, while every affordable bundle with both goods gives $u > 0$: the optimum is interior. Maximizing u is the same as maximizing $\ln u = \ln x_1 + 2 \ln x_2$, which is strictly concave, on the convex budget set: so the tangency conditions are sufficient and the solution is unique. At $p = (1, 1)$, $w = 6$: $x_1 = 6/3 = 2$ and $x_2 = 12/3 = 4$, so $x(p, w) = (2, 4)$.

(b) Indirect utility: utility at the demanded bundle,

$$v(p, w) = x_1(p, w)\, x_2(p, w)^2 = \frac{w}{3p_1} \cdot \frac{4w^2}{9p_2^2} \qquad \Longrightarrow \qquad v(p, w) = \frac{4w^3}{27\, p_1 p_2^2}.$$

At $p = (1, 1)$, $w = 6$: $v = 4 \cdot 216/27 = 32$ ($= 2 \cdot 4^2$, as it must be).

(c) Expenditure function. $e(p, u)$ is the wealth that just reaches utility u. Since v is strictly increasing in w, exactly one wealth level does: solve $v(p, e) = u$ for e (duality: $v(p, e(p, u)) = u$):

$$\frac{4e^3}{27 p_1 p_2^2} = u \implies e^3 = \frac{27}{4}\, p_1 p_2^2\, u \implies e(p, u) = 3\Big(\frac{p_1 p_2^2\, u}{4}\Big)^{1/3}.$$

Check: $e\big((1, 1), 32\big) = 3\,(32/4)^{1/3} = 3 \cdot 2 = 6 = w$.

(d) Hicksian demand by Shephard's lemma, $h_i(p, u) = \partial e(p, u)/\partial p_i$. (Why: $e(p, u) = p \cdot h(p, u)$; when p_i rises a little, the old cheapest bundle still reaches u and costs h_i more per euro of price rise, and re-optimizing the bundle changes the cost only to second order: the envelope theorem.) Write the expenditure function as a product of powers, $e(p, u) = 3\,(u/4)^{1/3}\, p_1^{1/3}\, p_2^{2/3}$. Differentiating a power, $\partial e/\partial p_1 = \frac{1}{3}\, e/p_1$ and $\partial e/\partial p_2 = \frac{2}{3}\, e/p_2$, so

$$h_1(p, u) = \frac{e(p, u)}{3p_1} = \frac{1}{p_1}\Big(\frac{p_1 p_2^2 u}{4}\Big)^{1/3}, \qquad h_2(p, u) = \frac{2\, e(p, u)}{3p_2} = \frac{2}{p_2}\Big(\frac{p_1 p_2^2 u}{4}\Big)^{1/3}.$$

Check $h(p, u) = x(p, e(p, u))$: by (a), $x_1(p, e) = \frac{e}{3p_1}$ and $x_2(p, e) = \frac{2e}{3p_2}$ with $e = e(p, u)$: exactly h_1 and h_2, for every p and u. (Duality: the cheapest way to reach u is what she demands with just enough wealth to reach it.) Also $u(h(p, u)) = \frac{e}{3p_1} \cdot \frac{4e^2}{9p_2^2} = \frac{4e^3}{27 p_1 p_2^2} = v(p, e) = u$ by (c).

(e) The price of good 2 rises: $p^0 = (1, 1) \to p^1 = (1, 8)$, $w = 6$.

- New bundle: $x_1(p^1, w) = \frac{6}{3} = 2$, $x_2(p^1, w) = \frac{12}{3 \cdot 8} = \frac{1}{2}$, so $x(p^1, w) = (2, \frac{1}{2})$; good 1 does not change (its share of wealth is fixed). Utilities: $u^0 = 32$ and $u^1 = 2 \cdot (\frac{1}{2})^2 = \frac{1}{2}$.
- Compensated bundle $h(p^1, u^0)$: at the new prices, give her just enough wealth to reach her old utility, $e(p^1, u^0) = 3\big(1 \cdot 64 \cdot 32/4\big)^{1/3} = 3 \cdot 512^{1/3} = 3 \cdot 8 = 24$ euros instead of 6; then $h_1(p^1, u^0) = \frac{24}{3} = 8$ and $h_2(p^1, u^0) = \frac{2 \cdot 24}{3 \cdot 8} = 2$ (check: $8 \cdot 2^2 = 32$, and it costs $8 + 16 = 24$).
- *Why the split goes through $h(p^1, u^0)$.* From $x(p^0, w)$ to $h(p^1, u^0)$ she stays on her old indifference curve u^0, so the move is a pure reaction to the new relative price: the substitution effect. From $h(p^1, u^0)$ to $x(p^1, w)$ the prices are already p^1 and only her wealth changes, from 24 back to $w = 6$ (the 18 euros of compensation are taken away): the wealth effect. Split of $\Delta x_2 = \frac{1}{2} - 4 = -\frac{7}{2}$:

  $$\text{substitution } h_2(p^1, u^0) - x_2(p^0, w) = 2 - 4 = -2\,, \quad \text{wealth } x_2(p^1, w) - h_2(p^1, u^0) = \tfrac{1}{2} - 2 = -\tfrac{3}{2}\,.$$

  Both are negative: the substitution effect of an own-price rise is never positive (the compensated law of demand), and good 2 is normal ($x_2 = 2w/(3p_2)$ rises with w), so the loss of wealth lowers it further. Good 1's Hicksian demand *rises*, from 2 to 8: compensated to stay on u^0, she substitutes towards good 1, which is now relatively cheaper (substitution effect on good 1: $8 - 2 = +6$); uncompensated, she keeps $x_1 = 2$ because the wealth effect, $2 - 8 = -6$, exactly offsets it (Cobb–Douglas: good 1's share of wealth does not depend on p_2).
- Welfare (MWG's signs: negative means a loss). $\mathrm{CV} = w - e(p^1, u^0) = 6 - 24 = -18$: at the new prices she would need 18 more euros to be as well off as before. $e(p^0, u^1) = 3\big(1 \cdot 1 \cdot \frac{1}{2}/4\big)^{1/3} = 3 \cdot (\frac{1}{8})^{1/3} = \frac{3}{2}$, so $\mathrm{EV} = e(p^0, u^1) - w = \frac{3}{2} - 6 = -\frac{9}{2}$: at the old prices, losing $\frac{9}{2}$ euros would hurt as much as the price rise.
  Why $|\mathrm{EV}| < |\mathrm{CV}|$. Since $w = e(p^0, u^0) = e(p^1, u^1)$, both are the cost of the same price rise at a fixed utility: $|\mathrm{CV}| = e(p^1, u^0) - e(p^0, u^0)$ at the old utility, $|\mathrm{EV}| = e(p^1, u^1) - e(p^0, u^1)$ at the new one. By Shephard's lemma $e(p^1, u) - e(p^0, u) = \int_1^8 h_2((1, t), u)\, dt$, the area to the left of good 2's Hicksian demand. Good 2 is normal, so h_2 is larger at the higher utility $u^0 = 32$ than at $u^1 = \frac{1}{2}$, at every price: the area at u^0 (18) is larger than the area at u^1 ($\frac{9}{2}$).

Problem 2 · $u(x_1, x_2) = 2\sqrt{x_1} + 3\sqrt{x_2}$

classes 4–7

(a) The utility maximization problem. Maximize $2\sqrt{x_1} + 3\sqrt{x_2}$ subject to $p_1 x_1 + p_2 x_2 \le w$. Since u is strictly increasing in each good, the budget binds at the optimum (otherwise a little more of either good would be affordable and better). With one multiplier λ for the budget constraint:

$$\mathcal{L} = 2\sqrt{x_1} + 3\sqrt{x_2} + \lambda\,(w - p_1 x_1 - p_2 x_2).$$

First-order conditions at an interior optimum:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \frac{1}{\sqrt{x_1}} - \lambda p_1 = 0, \qquad \frac{\partial \mathcal{L}}{\partial x_2} = \frac{3}{2\sqrt{x_2}} - \lambda p_2 = 0, \qquad p_1 x_1 + p_2 x_2 = w.$$

Divide the first two to eliminate λ: $\mathrm{MRS} = \frac{1/\sqrt{x_1}}{3/(2\sqrt{x_2})} = \frac{2}{3}\sqrt{\frac{x_2}{x_1}} = \frac{p_1}{p_2}$, so $\sqrt{x_2} = \frac{3p_1}{2p_2}\sqrt{x_1}$, i.e. $x_2 = \frac{9p_1^2}{4p_2^2}\, x_1$ (the tangency condition: the indifference curve touches the budget line). Put this into the budget: $p_1 x_1 + \frac{9p_1^2}{4p_2} x_1 = w$, i.e. $x_1\, \frac{p_1(4p_2 + 9p_1)}{4p_2} = w$, and $x_2 = \frac{9p_1^2}{4p_2^2}\, x_1$ gives x_2. Hence

$$x_1(p, w) = \frac{4w\, p_2}{p_1(4p_2 + 9p_1)}, \qquad x_2(p, w) = \frac{9w\, p_1}{p_2(4p_2 + 9p_1)}.$$

Why interior. The marginal utilities $\frac{1}{\sqrt{x_1}}$ and $\frac{3}{2\sqrt{x_2}}$ tend to $+\infty$ as $x_i \to 0$. At a bundle with $x_i = 0$, moving a little wealth into good i raises utility at an unboundedly high rate, more than it costs in the other good, so no corner bundle can be optimal: both goods are bought.
Why the first-order conditions are sufficient. Each $\sqrt{x_i}$ is concave, so u is concave (indeed strictly concave),

and the budget set is convex (the constraint is linear). For a concave objective on a convex set, a point that satisfies the first-order conditions (with $\lambda > 0$) is a global maximum; strict concavity makes it the only one. At $p = (1, 3)$, $w = 56$: $4p_2 + 9p_1 = 21$, so $x_1 = \frac{4 \cdot 56 \cdot 3}{1 \cdot 21} = 32$ and $x_2 = \frac{9 \cdot 56 \cdot 1}{3 \cdot 21} = 8$: $\boxed{x = (32, 8).}$

(b) Indirect utility. Put $x(p, w)$ into u and factor out $\sqrt{w/(4p_2 + 9p_1)}$:

$$v(p, w) = \sqrt{\frac{w}{4p_2 + 9p_1}}\left(2 \cdot 2\sqrt{\frac{p_2}{p_1}} + 3 \cdot 3\sqrt{\frac{p_1}{p_2}}\right) = \sqrt{\frac{w}{4p_2 + 9p_1}} \cdot \frac{4p_2 + 9p_1}{\sqrt{p_1 p_2}}$$

$$\Longrightarrow \qquad \boxed{v(p, w) = \sqrt{\frac{w\,(4p_2 + 9p_1)}{p_1\, p_2}}.}$$

(We used $4\sqrt{\frac{p_2}{p_1}} + 9\sqrt{\frac{p_1}{p_2}} = \frac{4p_2 + 9p_1}{\sqrt{p_1 p_2}}$, since $\sqrt{p_2/p_1} = p_2/\sqrt{p_1 p_2}$ and $\sqrt{p_1/p_2} = p_1/\sqrt{p_1 p_2}$.) Equivalently $v = \sqrt{w\,(4/p_1 + 9/p_2)}$. At $p = (1, 3)$, $w = 56$: $v^2 = 56 \cdot 21/3 = 392$, so $v = 14\sqrt{2}$ $(= 2\sqrt{32} + 3\sqrt{8} = 8\sqrt{2} + 6\sqrt{2})$. *A check by Roy's identity (not asked)*: differentiating $v^2 = w(4/p_1 + 9/p_2)$,

$$\frac{\partial v}{\partial w} = \frac{4/p_1 + 9/p_2}{2v}, \quad \frac{\partial v}{\partial p_1} = -\frac{4w}{2v\, p_1^2} \quad \Longrightarrow \quad -\frac{\partial v/\partial p_1}{\partial v/\partial w} = \frac{4w/p_1^2}{4/p_1 + 9/p_2} = \frac{4w\, p_2}{p_1(4p_2 + 9p_1)} = x_1(p, w).$$

(c) Expenditure function. As in P1(c), v is strictly increasing in w, so solve $v(p, e) = u$ for e: $u^2 = \frac{e\,(4p_2 + 9p_1)}{p_1 p_2}$, so

$$\boxed{e(p, u) = \frac{u^2\, p_1\, p_2}{4p_2 + 9p_1}.}$$

(d) Hicksian demand by Shephard's lemma (quotient rule):

$$h_1(p, u) = \frac{\partial e}{\partial p_1} = u^2\, \frac{p_2(4p_2 + 9p_1) - 9p_1 p_2}{(4p_2 + 9p_1)^2} \qquad \Longrightarrow \qquad \boxed{h_1(p, u) = \frac{4u^2\, p_2^2}{(4p_2 + 9p_1)^2},}$$

and h_2 in the same way, since $p_1(4p_2 + 9p_1) - 4p_1 p_2 = 9p_1^2$:

$$\boxed{h_2(p, u) = \frac{9u^2\, p_1^2}{(4p_2 + 9p_1)^2}.}$$

Check: $2\sqrt{h_1} + 3\sqrt{h_2} = \frac{4u\, p_2 + 9u\, p_1}{4p_2 + 9p_1} = u$, and $p \cdot h = \frac{u^2 p_1 p_2 (4p_2 + 9p_1)}{(4p_2 + 9p_1)^2} = e(p, u)$.

(e) The price rise $p^0 = (1, 3) \to p^1 = (8, 3)$, $w = 56$. Tip: work with the square of utility, which is a whole number here.

- New bundle: $4p_2 + 9p_1 = 12 + 72 = 84$, so $x_1 = \frac{4 \cdot 56 \cdot 3}{8 \cdot 84} = 1$ and $x_2 = \frac{9 \cdot 56 \cdot 8}{3 \cdot 84} = 16$: $\boxed{x(p^1, w) = (1, 16)}$ and $\boxed{u^1 = 2\sqrt{1} + 3\sqrt{16} = 14.}$ Before: $u^0 = 14\sqrt{2}$, so $(u^0)^2 = 392$.
- $e(p^1, u^0) = \frac{392 \cdot 8 \cdot 3}{84} = 112$, so $\mathrm{CV} = w - e(p^1, u^0) = 56 - 112 = \boxed{-56.}$ At the new prices she would need her wealth doubled $(112 = 2w)$ to be as well off as before.
- $e(p^0, u^1) = \frac{196 \cdot 1 \cdot 3}{21} = 28$, so $\mathrm{EV} = e(p^0, u^1) - w = 28 - 56 = \boxed{-28.}$ At the old prices, halving her wealth $(28 = w/2)$ would hurt as much as the price rise. $|\mathrm{EV}| < |\mathrm{CV}|$ for the same reason as in P1(e): good 1 is normal (x_1 rises with w).
- $h_1(p^1, u^0) = \frac{4 \cdot 392 \cdot 9}{84^2} = \frac{392}{84} \cdot \frac{36}{84} = \frac{14}{3} \cdot \frac{3}{7} = 2$ (and $h_2 = \frac{9 \cdot 392 \cdot 64}{84^2} = 32$; check: $8 \cdot 2 + 3 \cdot 32 = 112 = e(p^1, u^0)$). A shortcut without the big products: expenditure minimization has the same tangency condition $\mathrm{MRS} = p_1/p_2$, so $h(p^1, u^0)$ lies on the same line $\sqrt{x_2} = \frac{3p_1}{2p_2}\sqrt{x_1}$ as the demand at p^1, at the point that reaches u^0. At p^1, $\sqrt{x_2} = \frac{3p_1}{2p_2}\sqrt{x_1} = 4\sqrt{x_1}$, so $u = 2\sqrt{x_1} + 12\sqrt{x_1} = 14\sqrt{x_1}$; setting it equal to $u^0 = 14\sqrt{2}$ gives $x_1 = 2$ and $x_2 = 16 \cdot 2 = 32$. Split of $\Delta x_1 = 1 - 32 = -31$:

  substitution $h_1(p^1, u^0) - x_1(p^0, w) = 2 - 32 = \boxed{-30}$, wealth $x_1(p^1, w) - h_1(p^1, u^0) = 1 - 2 = \boxed{-1}$.

  Most of the fall in x_1 is substitution: the goods substitute easily, and x_2 rises from 8 to 16 (gross substitutes).

Problem 3 · Expected utility: a two-stage game

class 8

(a) Reduced lottery. Multiply along each branch of the tree: the die gives lottery A with probability $\frac{2}{6}=\frac{1}{3}$ and lottery B with probability $\frac{2}{3}$.

$$144:\ \tfrac{1}{3}\cdot\tfrac{1}{4}=\tfrac{1}{12},\qquad 0:\ \tfrac{1}{3}\cdot\tfrac{3}{4}=\tfrac{1}{4},\qquad 81:\ \tfrac{2}{3}\cdot\tfrac{1}{2}=\tfrac{1}{3},\qquad 9:\ \tfrac{2}{3}\cdot\tfrac{1}{2}=\tfrac{1}{3}.$$

prizes $(0,9,81,144)$ euros with probabilities $(\frac{1}{4},\frac{1}{3},\frac{1}{3},\frac{1}{12})$. Check: $\frac{3+4+4+1}{12}=1$. For an expected-utility maximizer only this reduced lottery matters.

(b) Expected value, expected utility, certainty equivalent, risk premium.

$$E[x]=\tfrac{1}{4}\cdot 0+\tfrac{1}{3}\cdot 9+\tfrac{1}{3}\cdot 81+\tfrac{1}{12}\cdot 144=0+3+27+12=\ 42 \text{ euros},$$

$$E[u(x)]=\tfrac{1}{4}\cdot 0+\tfrac{1}{3}\cdot 3+\tfrac{1}{3}\cdot 9+\tfrac{1}{12}\cdot 12=0+1+3+1=\ 5 \text{ utils}.$$

The certainty equivalent solves $\sqrt{c}=5$: $c=25$ euros, so $\mathrm{RP}=E[x]-c=42-25=17$ euros. *Attitude to risk.* $u(x)=\sqrt{x}$ is strictly concave ($u''=-\frac{1}{4}x^{-3/2}<0$), so Martina is risk averse: by Jensen's inequality $E[u(x)]\le u(E[x])$, strictly for a risky lottery (here $5<\sqrt{42}$). Since u is increasing, $u(c)=E[u(x)]<u(E[x])$ means that her certainty equivalent lies below the mean, $c=25<42=E[x]$, and the risk premium is positive: she would give up 17 euros of expected value to remove the risk.

(c) The offer: s euros for sure in place of B. Both options are compound lotteries that share the branch "lottery A with probability $\frac{1}{3}$":

$$\text{keep: } \tfrac{1}{3}A+\tfrac{2}{3}B,\qquad \text{offer: } \tfrac{1}{3}A+\tfrac{2}{3}\,s.$$

The *independence axiom* says: for all lotteries L,L',L'' and $\alpha\in(0,1)$, $L\succsim L'\iff \alpha L+(1-\alpha)L''\succsim \alpha L'+(1-\alpha)L''$. Take $L=s$ (the sure amount), $L'=B$, $L''=A$ and $\alpha=\frac{2}{3}$: she weakly prefers the offer if and only if she weakly prefers s for sure to B, whatever lottery A is. So A does not matter. Now compare s with B: $E_B[u]=\frac{1}{2}\sqrt{81}+\frac{1}{2}\sqrt{9}=\frac{9+3}{2}=6$, so the certainty equivalent of B is $c_B=6^2=36$ euros, and

$$s\succsim B\iff \sqrt{s}\ge 6\iff s\ge 36 \text{ euros}.$$

(Check with the whole game: $\frac{1}{3}E_A[u]+\frac{2}{3}\sqrt{s}\ge\frac{1}{3}E_A[u]+\frac{2}{3}\cdot 6$, and the A term, $\frac{1}{3}\cdot 3$, cancels.) At $s=36$ she is indifferent.

(d) Pau, risk neutral: his Bernoulli utility is linear, so he ranks lotteries by their expected value and his certainty equivalent of B is its mean, $E_B[x]=\frac{81+9}{2}=45$. By the same independence argument $s\ge 45$ euros. A **strictly risk-loving** person (strictly convex u) has $E[u(x)]>u(E[x])$ for a risky lottery, so her certainty equivalent of B is *above* its mean: she needs a larger s than Pau, more than 45 euros. The ranking of the three thresholds is risk averse $36<$ risk neutral $45<$ risk loving.

Problem 4 (optional) · $u(x_1,x_2)=x_1+\min\{x_1,x_2\}$ on $\mathbb{R}^2_+$

class 2

(a) Monotone: yes. Let $y\gg x$. Then $y_1>x_1$ and $y_2>x_2$, so $\min\{y_1,y_2\}>\min\{x_1,x_2\}$ as well (each of y_1,y_2 exceeds $\min\{x_1,x_2\}$). Hence $u(y)-u(x)=(y_1-x_1)+(\min\{y_1,y_2\}-\min\{x_1,x_2\})>0$: $y\succ x$.

(b) Strongly monotone: no. Take $x=(1,2)$ and $y=(1,3)$: $y\ge x$ and $y\ne x$, but $u(y)=1+\min\{1,3\}=2=1+\min\{1,2\}=u(x)$, so $y\sim x$, not $y\succ x$. (Good 2 beyond x_1 units adds nothing.)

(c) Locally non-satiated: yes. Take any x and $\varepsilon>0$; the bundle $y=(x_1+\varepsilon/2,\ x_2)$ is within distance ε of x, lies in $\mathbb{R}^2_+$, and $u(y)\ge u(x)+\varepsilon/2>u(x)$ (the first term rises by $\varepsilon/2$ and the minimum does not fall). Or: monotone preferences are always LNS.

(d) Convex: yes. $u(x)=x_1+\min\{x_1,x_2\}=\min\{2x_1,\ x_1+x_2\}$, the minimum of two linear functions, so u is concave (at a mixture of two bundles, each linear function equals the mixture of its values, which is at least the mixture of the minima). Then for any x, if $u(y)\ge u(x)$ and $u(z)\ge u(x)$, for $\alpha\in[0,1]$: $u(\alpha y+(1-\alpha)z)\ge\alpha u(y)+(1-\alpha)u(z)\ge u(x)$. So every upper contour set $\{y: y\succsim x\}$ is convex. (Not strictly convex: the indifference curves are made of straight pieces, a vertical one where $x_2\ge x_1$.)

Every number in these solutions was checked with exact fractions. Questions: pkocourek@ub.edu.